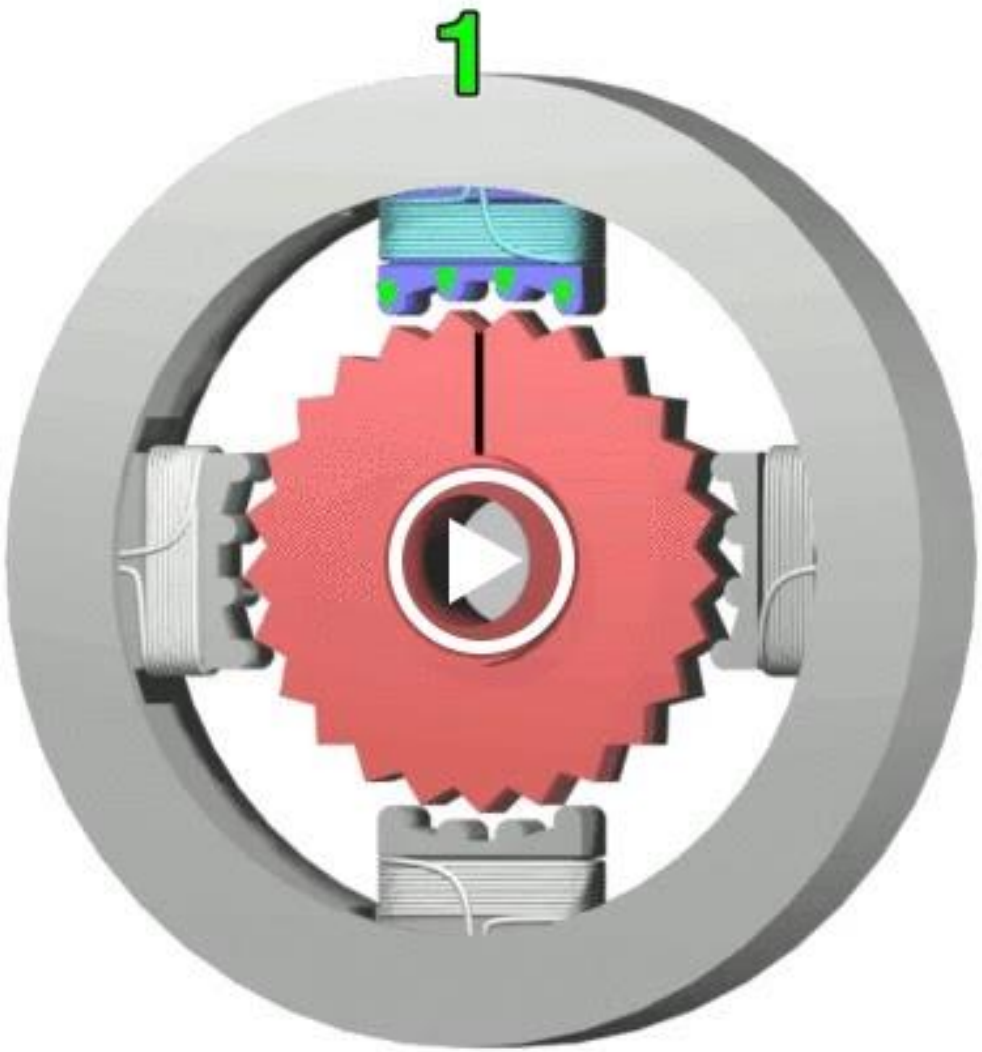
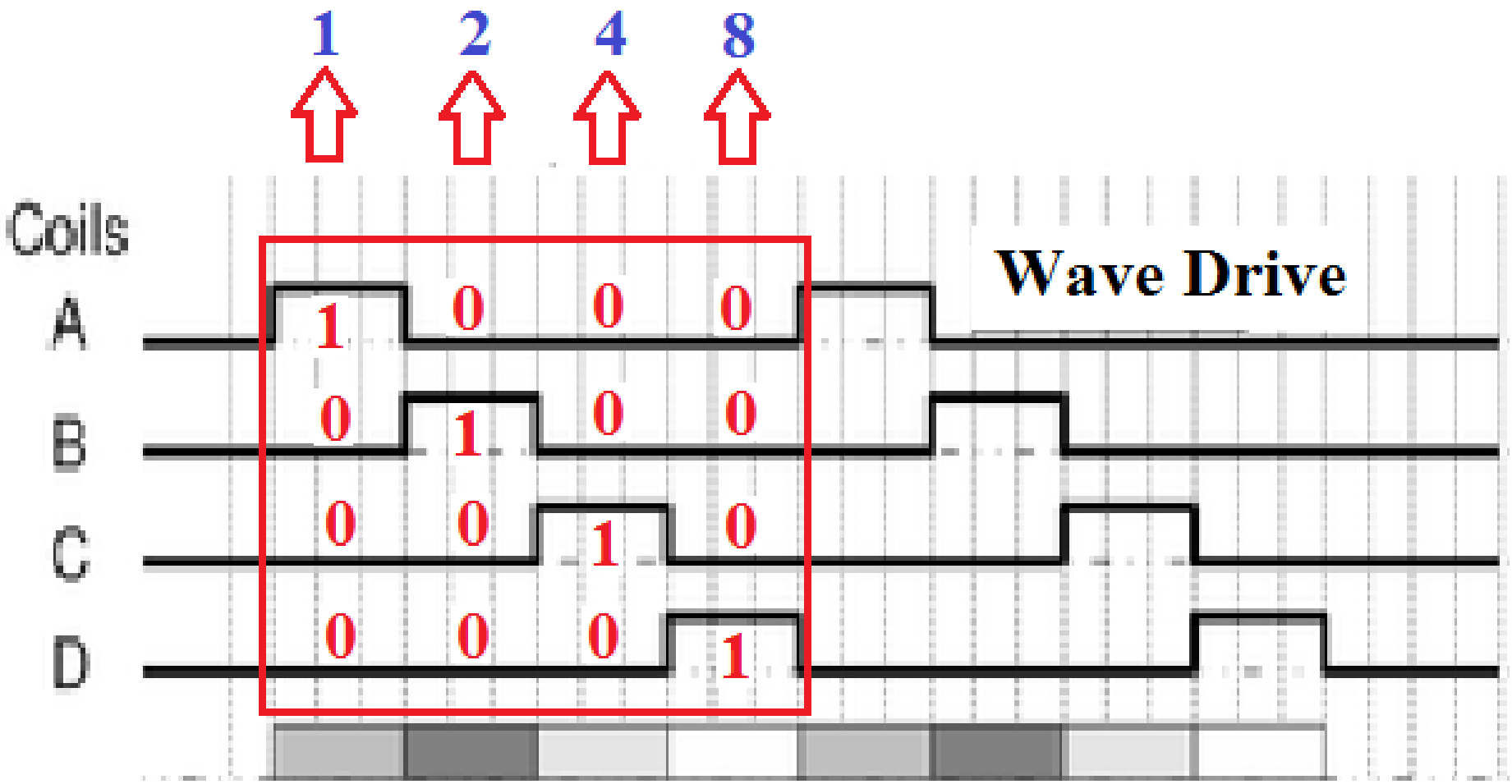
Stepper Motor Interfacing

Stepper motors are the motors that move in discrete steps or convert electrical pulses into rotatory motion. They have multiple coils(4coils) that are organized in groups called "phases"(stators named as A,B,C and D). By energizing each phase in sequence, the motor will rotate, one step at a time.


1



1 step sequence --> 1.8' rotation

4 such step sequence --> 7.2' rotation

Count value for 360' rotation is 360'/7.2'=50

Stepper Motor Interface with LPC1768

```
#include <LPC17xx.H>
void clock_wise(void);
void anti_clock_wise(void);
unsigned long int var1;
unsigned int i=0,j=0,k=0;
int main(void)  {
        LPC_PINCON->PINSEL0 = 0xFFFF00FF; //P0.4 to P0.7 GPIo
        LPC_GPIO0->FIODIR = 0x000000F0; //P0.4 to P0.7 output
         while(1)  {   for(j=0;j<50;j++)
                        clock_wise();
                 for(k=0;k<65000;k++);
                 for(j=0;j<50;j++)
                        anti_clock_wise();
                for(k=0;k<65000;k++);
    }}
```

Stepper Motor Interface with LPC1768

```
void clock_wise(void)
{
        var1 = 0x00000010;      //For Clockwise
         for(i=0;i<=3;i++)      // for A B C D Stepping
        {
                LPC_GPIO0->FIOPIN = var1;
                var1 = var1<<1;      //For Clockwise
                for(k=0;k<3000;k++); //for step speed variation
         }

}
```

Stepper Motor Interface with LPC1768

```
void anti_clock_wise(void)
{
        var1 = 0x00000100;        //For Anticlockwise
         for(i=0;i<=3;i++)        // for  D C B A Stepping
        {
                var1 = var1>>1;        //For Anticlockwise
                LPC_GPIO0->FIOPIN = var1;
                for(k=0;k<3000;k++); //for step speed variation
         }

}
```